Rural Policy Centre Research Briefing

SRUC

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Incineration and Energy from Waste[1]

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Key message: Incineration[3] and Energy from Waste (EfW) provides a more sustainable alternative to landfill disposal by converting waste into usable energy. However, potential health and environmental impacts, as well as the risk of undermining higher-priority waste management practices like reuse and recycling, must be carefully considered.

Main Findings

- This document aims to provide an overview of waste incineration in EfW facilities in Scotland. It presents information on waste trends and regulations concerning EfW, highlights recent policy developments within the framework of Scotland's transition to a circular economy.
- The Scottish Parliament (SPICe) analysed existing literature on EfW technology, relevant policy documents, and emissions data.
- The upcoming ban on biodegradable municipal waste (BMW) in 2025, along with rising landfill tax, has contributed to an increase in incineration as waste is redirected from landfills to EfW facilities.
- Plans for new incineration facilities face delays pending a review from the Scottish Government. The review will examine emission reduction, heat reuse, health impacts, and how incineration fits with Scotland's waste hierarchy and other treatment options.
- The draft National Planning Framework 4 (NPF4) will establish new policies for EfW and will replace the current Scottish Planning Policy (SPP).

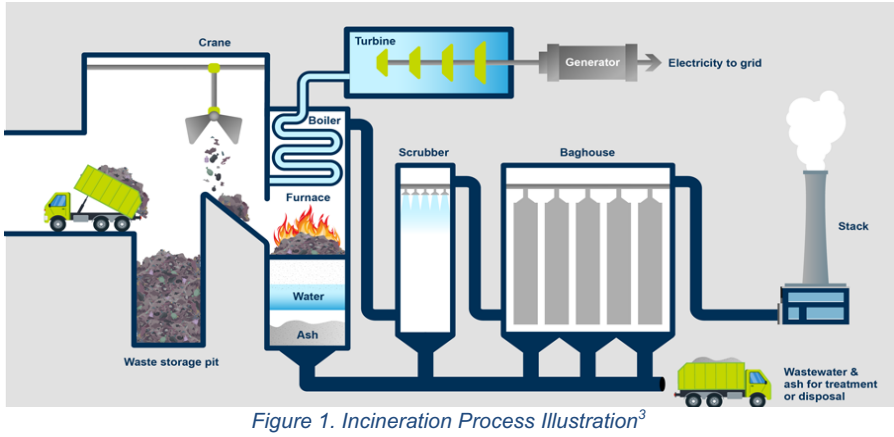

Figure 1. Incineration Process Illustration[3]

[1] This research was conducted by the Scottish Parliament Information Center (SPICe), SB 22-06, February 2022.
[2] *Scottish Parliament Information Centre (SPICe) Researchers.* The authors are unable to engage in individual discussions regarding SPICe Briefing Papers. For comments, please contact SPICe at: SPICe@parliament.scot. For additional inquiries, reach out to the Parliament's Public Information Service at: sp.info@parliament.scot.
[3] For further information on incineration please visit: https://www.futurelearn.com/info/courses/introduction-to-environmental-science/0/steps/271560.

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Introduction

The Scottish Government will implement a legal ban on biodegradable municipal waste being sent to landfill starting in 2025. In response, local authorities have planned to build more EfW facilities, rising concerns among the public and experts.

There is a dilemma between ensuring there is enough incineration capacity by 2025 to comply with the ban and avoiding the construction of too many incinerators. If too many incinerators are built now, it could result in an overreliance on incineration, which might distract from efforts to focus on recycling and reducing waste.

Methods

Limitations of the study include a lack of specific data and significant variances in regional data.

Results

In 2022, Scotland had six operational Energy from Waste (EfW) sites for municipal solid waste, with a combined annual capacity of 1.1 million tonnes. Additionally, there are plans for 13 new incinerators that will provide an extra 2.5 million tonnes of capacity (See Figure 2).

This potentially result in overcapacity for incineration in the coming years, especially considering that only 0.7 million tonnes of biodegradable municipal waste (BMW) went to landfill in 2020 (See Figure 3).

This surplus capacity could lead to unnecessary resource waste by incinerating materials that could be recycled, particularly as Scotland intensifies efforts to meet recycling targets, which would further decrease the volume of BMW in the future.

Some experts warned that constructing new incinerators would cause a reliance on burning resources, causing further environmental harm. To address the current shortfall in waste treatment capacity while avoiding overreliance on incineration, the Scottish Government should swiftly conclude its EfW review to promote investment in the necessary waste treatment infrastructure.

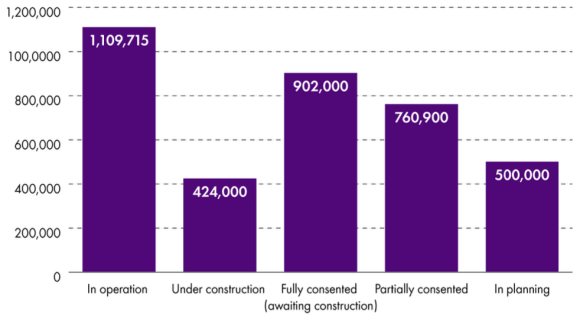

Figure 2. Waste capacity (tonnes MSW per year) of EfW facilities in Scotland. [Source: SEPA, 2021]

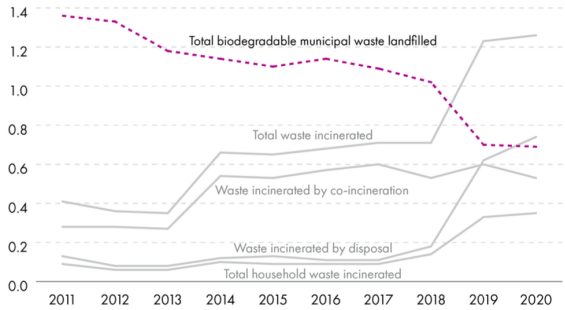

Figure 3. Waste incinerated in Scotland 2011-2020 (in million tonnes) [Source: SEPA, 2021]

Policy Implications

To ensure effective waste management in Scotland, policies should align with the waste hierarchy by prioritizing waste reduction and recycling while treating incineration as a last resort. There must be robust monitoring of emissions from EfW facilities to safeguard public health and the environment, alongside infrastructure development to utilize heat recovery and carbon capture.

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